



New combined Inverse-QSAR and molecular docking method for scaffold-based drug discovery

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ABSTRACT

Computer-aided drug discovery plays a vital role in developing novel medications for various diseases. The COVID-19 pandemic has heightened the need for innovative approaches to design lead compounds with the potential to become effective drugs. Specifically, designing promising inhibitors of the SARS-CoV-2 main protease (Mpro) is crucial, as it plays a key role in viral replication. Phytochemicals, primarily flavonoids and flavonols from medicinal plants, were screened. Fifty small molecules were selected for molecular docking analysis against SARS-CoV-2 Mpro (PDB ID: 6LU7). Binding energies and interactions were analyzed and compared to those of the anti-SARS-CoV-2 inhibitor Nirmatrelvir. Using these 50 structures as a training set, a QSAR model was built employing simple, reversible topological descriptors. An inverse-QSAR analysis was then performed on $2^9 = 512$ hydroxyl combinations at nine possible positions on the flavone and flavonol scaffold. The model predicted three novel, promising compounds exhibiting the most favorable binding energies (-8.5 kcal/mol) among the 512 possible hydroxyl combinations: 3,6,7,2',4'-pentahydroxyflavone (PF9), 6,7,2',4'-tetrahydroxyflavone (PF11), and 3,6,7,4'-tetrahydroxyflavone (PF15). Molecular dynamics (MD) simulations demonstrated the stability of the PF9/Mpro complex over 300 ns of simulation. These predicted structures, reported here for the first time, warrant synthesis and further evaluation of their biological activity through in vitro and in vivo studies.

1. Introduction

Drug discovery is crucial for developing new treatments for a wide range of diseases [1]. The emergence of viruses like SARS-CoV-2 has underscored the critical need for efficient drug discovery methods and techniques [2]. Traditional experimental approaches, often reliant on laborious lab work and trial-and-error, have proven slow and expensive [2,3]. However, computer-aided drug discovery (CADD) is revolutionizing the field with novel computational methods [4]. CADD offers a powerful toolkit, including molecular docking, molecular dynamics (MD) simulations, quantitative structure-activity relationship (QSAR) modeling, and pharmacokinetic profiling. These tools enable researchers to analyze vast datasets, predict potential drug candidates, optimize their properties, and ultimately accelerate the discovery process [5]. These computational methods go beyond simple screening, offering profound insights into drug action mechanisms. For example, recent studies have employed molecular docking and MD simulations to elucidate the dual-targeting mechanism of serotonin-norepinephrine

reuptake inhibitors [6], showcasing the power of these methods in identifying key residues and structural determinants crucial for drug binding and selectivity. Similarly, MD simulations have been employed to unravel the allosteric inhibition of the serotonin transporter by escitalopram [7], highlighting the potential for designing novel drugs that target allosteric sites. Moreover, recognizing that even powerful machine learning-based structure prediction methods like AlphaFold2 have limitations in accurately modeling protein variants, researchers have developed novel strategies like MoDAFold [8]. This approach combines AlphaFold2 with MD simulations to enhance the accuracy of protein structure prediction, especially for variants with missense mutations. These advances are particularly relevant for rapidly evolving viruses like SARS-CoV-2, as understanding the structural impact of mutations on viral proteins is crucial for designing effective and resilient therapies.

In addition to targeting viral proteins, CADD is being harnessed to explore the rich molecular diversity of plants for novel drug candidates. Recent investigations have highlighted the potential of CADD for identifying promising molecules from natural sources. For instance, one

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study identified syringaldehyde as a potential inhibitor of the tubulin-colchicine receptor using molecular docking and MD simulations to investigate the anthelmintic activity of pineapple compounds [9]. Similarly, molecular docking was employed to assess the analgesic potential of *Erythrina variegata* L. derived phytochemicals against COX enzymes, revealing phaseollin as a promising compound with high binding affinity [10]. In another study, researchers explored the antifungal activity of uridine derivatives using a combination of in silico techniques, including ADME/T analysis, PASS prediction, molecular docking, and MM-GBSA calculations [11]. Methyl β -D-galactopyranoside analogs were also investigated for their antimicrobial and potential antiviral properties using an array of computational methods, including molecular docking, MD simulations, and pharmacokinetic predictions [12]. Furthermore, researchers scrutinized the antidepressant and cytotoxic activities of compounds from *Cycas pectinata* Buch.-Ham. using a synergistic blend of biological evaluations, DFT calculations, and molecular docking studies [13]. The anxiolytic and antidepressant potential of *Cnesmone javanica* Blume leaf extract was explored using network pharmacology to predict relevant pathways and molecular docking to assess binding against serotonin transporter and potassium channel receptors [14]. Lastly, researchers investigated the antimicrobial efficacy of *Bacopa monnieri* leaf extract against *Staphylococcus aureus* using in vitro assays and molecular docking studies. These studies identified luteolin, a potent antimicrobial compound, by targeting penicillin-binding protein and DNA gyrase [15]. Taken together, these studies illustrate the remarkable versatility of CADD as a powerful tool for unlocking the therapeutic potential of plant-derived molecules.

Phytochemical scaffolds hold promise as SARS-CoV-2 inhibitors and provide valuable leads for drug development [16]. Polyphenols, a class

of phytochemicals, primarily include flavonoids, tannins, stilbenes, coumarins, and phenolic acids [17]. (Fig. 1). These compounds are abundant in common foods, such as fruits, vegetables, and nuts [18]. Evidence increasingly points to their role in preventing degenerative diseases and their positive effects on human health. Polyphenols have been extensively studied for their antioxidant properties, specifically their ability to scavenge harmful reactive oxygen species (ROS) [18,19]. Furthermore, these phytochemicals exhibit a broad spectrum of biological activities, including anticancer [20], antimicrobial [20–22], antiviral properties [23,24]. Their inhibitory potential against coronaviruses, including SARS-CoV [25,26], SARS-CoV [27], MERS-CoV [25] and SARS-CoV-2 [28,29], has been of particular interest. Specifically, the antiviral activity of flavonols [30] and flavones [31] against SARS-CoV-2 has been investigated using molecular docking [32–36] and molecular dynamics simulations [36–38] primarily targeting the SARS-CoV-2 Mpro protein, which is essential for viral replication and proliferation [39].

Designing novel compounds (*de novo* design) with favorable pharmacological properties remains a significant challenge in cheminformatics and computational chemistry. Inverse-QSAR techniques, which generate new molecular structures based on established QSAR models, offer a promising approach for *de novo* design [40,41]. However, despite their potential, inverse-QSAR techniques are underutilized, primarily due to the challenge of deriving molecular structures from conventional QSAR descriptors. This hurdle necessitates the development of novel, reversible descriptors that can facilitate the construction of new molecules with desired activities [42].

Several studies, including those by Safarizadeh and Garkani-Nejad (2019) [43] and Tuffaha, Hatmal, and Taha (2019) [44], have explored combining QSAR and molecular docking to model calculated binding energy. These models use multiple linear regression (MLR) to correlate binding energy with various molecular descriptors of selected compounds. Researchers have applied this combined QSAR and molecular docking approach to various targets related to SARS-CoV-2. For instance, Islam et al. (2021) [34] and Achutha, Pushpa, and Suchitra (2020) [45] focused on SARS-CoV-2 Mpro, while Basu, Sarkar, and Maulik (2020) [46] modeled the ACE2-spike protein interaction. Mamun et al. (2021) [47] investigated the SARS-CoV-2 spike protein. Beyond MLR, other modeling techniques have been explored, including partial least squares (PLS) by other researchers, like Ahmed et al. (2021) [48]. Machine learning strategies have been employed by Tejera et al. (2020) [49] and a combination of Genetic Algorithm (GA) and Artificial Neural Network (ANN) by Gholivand et al. (2022) [50]. To our knowledge, this is the first study to utilize these combined computational methods for the *de novo* design, moving beyond previous applications in virtual screening, such as those by Achutha, Pushpa, and Suchitra (2020) [45], Tejera et al. (2020) [49] and other researchers [51,52], were used for database virtual screening.

This study introduces a novel *de novo* drug design method integrating molecular docking and inverse-QSAR, with its efficacy showcased through the design of SARS-CoV-2 Mpro inhibitors. By exploring $2^9 = 512$ hydroxyl combinations on the flavone and flavonol scaffold, this study identifies promising lead compounds [53,54] for further experimental validation. While demonstrated here for SARS-CoV-2 Mpro, this adaptable method holds potential for broader application in drug discovery targeting diverse therapeutic targets.

2. Methods and materials

2.1. Data curation and preparation

Fifty flavonol and flavone compounds (Table 1) were retrieved from the ChEMBL database [55]. Each compound's structure was pre-optimized using the steepest descent algorithm with a convergence criterion of 10^{-6} kJ.mol $^{-1}$ at the MMFF94s force field. Following a conformational search, the lowest-energy conformer was further

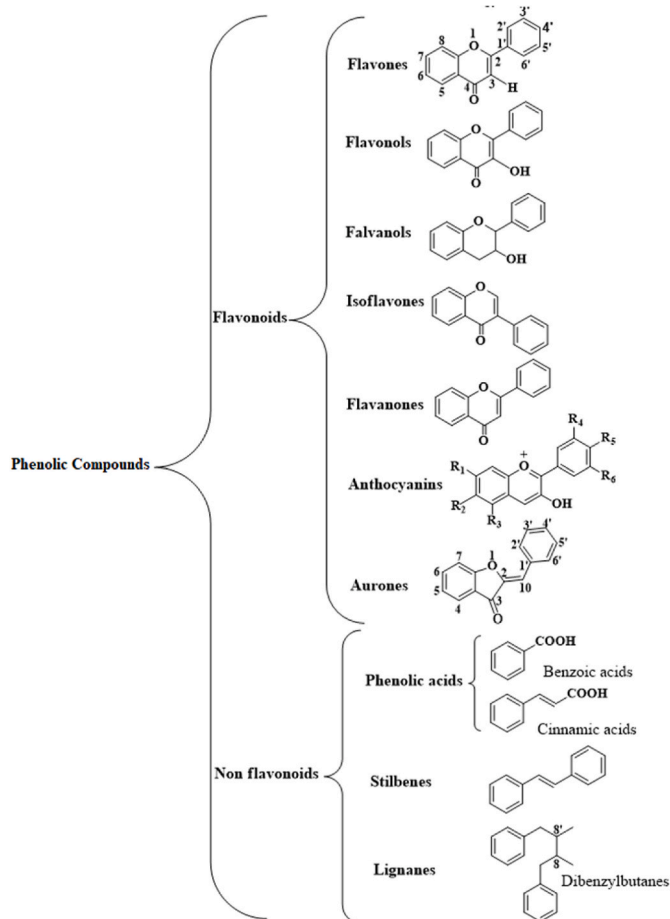


Fig. 1. The flavone basic structure and the main classes of polyphenols.

Table 1

The 50 structures ranked by AutoDock Vina Docking Score with SARS-CoV-2 Mpro.

Molecule	ChEMBL ID	R3	R5	R6	R7	R8	R2'	R3'	R4'	R5'	Rank	Binding Energy Kcal/mol
Nirmatrelvir	4802135										Ref 1	−8.7
5,6,7,4'-Tetrahydroxyflavone (6-hydroxyapigenin)	55415	0	1	1	1	0	0	0	1	0	F1	−8.3
3,6,3',4'-Tetrahydroxyflavone	1224325	1	0	1	0	0	0	1	1	0	F2	−8.2
6,3',4'-Trihydroxyflavone	1224323	0	0	1	0	0	0	1	1	0	F3	−8.2
6,7,3',4'-Tetrahydroxyflavone	1224404	0	0	1	1	0	0	1	1	0	F4	−8.2
7,4'-Dihydroxyflavone	294878	0	0	0	1	0	0	0	1	0	F5	−8.1
3,6,2',4',5'-Pentahydroxyflavone	1224493	1	0	1	0	0	1	0	1	1	F6	−8.1
3,5,7,4'-Tetrahydroxyflavone (kaempferol)	150	1	1	0	1	0	0	0	1	0	F7	−8
3,7,3',4'-Tetrahydroxyflavone (fisetin)	31574	1	0	0	1	0	0	1	1	0	F8	−8
3,5,7,2',4'-Pentahydroxyflavone (morin)	28626	1	1	0	1	0	1	0	1	0	F9	−8
3,5,7,3',4'-Pentahydroxyflavone (quercetin)	31574	1	1	0	1	0	0	1	1	0	F10	−7.9
7,3',4'-Trihydroxyflavone	301624	0	0	0	1	0	0	1	1	0	F11	−7.9
6,7,3'-Trihydroxyflavone	1801016	0	0	1	1	0	0	1	0	0	F12	−7.9
3,6,2',4'-Tetrahydroxyflavone	400890	1	0	1	0	0	1	0	1	0	F13	−7.9
5,7,3',4'-Tetrahydroxyflavone (Luteolin)	151	0	1	0	1	0	0	1	1	0	F14	−7.8
3',4'-Dihydroxyflavone	222556	0	0	0	0	0	0	1	1	0	F15	−7.8
3,3',4'-Trihydroxyflavone	122701	1	0	0	0	0	0	1	1	0	F16	−7.8
5,3',4'-Trihydroxyflavone	243677	0	1	0	0	0	0	1	1	0	F17	−7.7
6,2',3'-Trihydroxyflavone	1224322	0	0	1	0	0	1	1	0	0	F18	−7.7
3,5,7,8,3',4'-Hexahydroxyflavone (gossypetin)	253570	1	1	0	1	1	0	1	1	0	F19	−7.7
3,7,3',4',5'-Pentahydroxyflavone	170405	1	0	0	1	0	0	1	1	1	F20	−7.6
7,2'-Dihydroxyflavone	475908	0	0	0	1	0	1	0	0	0	F21	−7.6
7,3',4',5'-Tetrahydroxyflavone	398328	0	0	0	1	0	0	1	1	1	F22	−7.6
3,6,2',3'-Tetrahydroxyflavone	1224324	1	0	1	0	0	1	1	0	0	F23	−7.6
7,3'-Dihydroxyflavone	253786	0	0	0	1	0	0	1	0	0	F24	−7.5
3,5,7,3',4',5'-Hexahydroxyflavone (myricetin)	164	1	1	0	1	0	0	1	1	1	F25	−7.5
5,7,2'-Trihydroxyflavone	242385	0	1	0	1	0	1	0	0	0	F26	−7.5
3,5,7,2'-Tetrahydroxyflavone (datiscetin)	503168	1	1	0	1	0	1	0	0	0	F27	−7.5
5,7,3',4',5'-Pentahydroxyflavone	247484	0	1	0	1	0	0	1	1	1	F28	−7.5
3,5,7-Trihydroxyflavone (galangin)	309490	1	1	0	1	0	0	0	0	0	F29	−7.5
5,7-Dihydroxyflavone (Chrysin)	117	0	1	0	1	0	0	0	0	0	F30	−7.5
2',3'-Dihydroxyflavone	242171	0	0	0	0	0	1	1	0	0	F31	−7.5
7,8,3',4'-Tetrahydroxyflavone	222541	0	0	0	1	1	0	1	1	0	F32	−7.5
7,8,4'-Trihydroxyflavone	420982	0	0	0	1	1	0	0	1	0	F33	−7.4
2',4'-Dihydroxyflavone	1374610	0	0	0	0	0	1	0	1	0	F34	−7.4
7,8,2'-Trihydroxyflavone	244707	0	0	0	1	1	1	0	0	0	F35	−7.4
7,8,3'-Trihydroxyflavone	225513	0	0	0	1	1	0	1	0	0	F36	−7.4
3,6,2'-Trihydroxyflavone	1224564	1	0	1	0	0	1	0	0	0	F37	−7.4
3',5'-Dihydroxyflavone	1224564	0	0	0	0	0	0	1	0	1	F38	−7.3
3'-Hydroxyflavone	147552	0	0	0	0	0	0	1	0	0	F39	−7.3
4'-Hydroxyflavone	327209	0	0	0	0	0	0	0	1	0	F40	−7.3
5,7,8-Trihydroxyflavone	485250	0	1	0	1	1	0	0	0	0	F41	−7.3
5,4'-Dihydroxyflavone	484662	0	1	0	0	0	0	0	1	0	F42	−7.3
6-Hydroxyflavone	138649	0	0	1	0	0	0	0	0	0	F43	−7.3
3,7,8,2'-Tetrahydroxyflavone	1224403	1	0	0	1	1	1	0	0	0	F44	−7.3
5,3'-Dihydroxyflavone	502242	0	1	0	0	0	0	1	0	0	F45	−7.2
5-Hydroxyflavone (Primuletin)	16807	0	1	0	0	0	0	0	0	0	F46	−7.1
Flavone	275638	0	0	0	0	0	0	0	0	0	F47	−7
3,2'-Dihydroxyflavone	1224243	1	0	0	0	0	1	0	0	0	F48	−7
2'-Hydroxyflavone	144278	0	0	0	0	0	1	0	0	0	F49	−6.8
3,5-Dihydroxyflavone	239276	1	1	0	0	0	0	0	0	0	F50	−6.8

optimized using the same force field as implemented in Avogadro software [56]. A final optimization was performed using MOPAC software at the PM7 theory level [57]. The SARS-CoV-2 main protease (Mpro) (PDB ID: 6LU7) [58], also known as chymotrypsin-like protease (3CLpro), was chosen as the target due to its direct role in the replication of coronavirus [39].

2.2. Molecular docking

The docking of each optimized structure against Mpro was performed using AutoDock Vina [59] as implemented in the COVID-19 Docking Server [60]. The server utilizes Mpro, co-crystallized with the inhibitor N3, downloaded from the PDB database (PDB ID: 6LU7). The docking box was centered on the native ligand with dimensions of 30 Å × 30 Å × 30 Å to include the residues of the entire cavity. OpenBabel was used for format transformation and 3D coordinate generation for the uploaded files [61]. The visualization of the results was carried out using Discovery Studio Visualizer software [62].

2.3. Molecular dynamics simulations

The molecular dynamics (MD) simulations were performed using GROMACS 2021.7 [63] to investigate the structural properties and stability. The MD simulation was carried out to PF9/6LU7 complex, having the best binding energy from molecular docking study. The system was hydrogens added and solvated in a cubic box with a 2 Å margin using an explicit water model TIP3P and the Amber99SB-ILDN force field. Ions were added proportionally to neutralize the system and simulate the physiological conditions at a concentration of 0.2 M. Specifically, Na⁺ and K⁺ ions were used as cations, in addition to Cl[−] ions for anionic effect. An energy optimization of the whole system was performed in two steps: 5000 steps using the steepest descent method then 5000 steps using the conjugate gradient method. Both optimizations were done without position constraints. The Generalized Amber Force Field (GAFF) was used for PF9 ligand, while the Amber99SB-ILDN force field was used for proteins. van der Waals interactions were treated with a cutoff distance of 10 Å.

After applying the MD cycle with the canonical NVT environment set at 1 ns, the system was kept at constant temperature (298.15 K). Subsequently, the NPT environment was established as the production phase, applying an MD cycle at 298.15 K and for 300 ns. The simulation time interval was set to 2 **femtoseconds** (fs), and periodic boundary conditions were applied. The MD results have been plotted by matplotlib package under python Jupyter notebook.

2.4. QSAR

The QSAR modeling was conducted using the QSARINS software [64]. For the modeling process, simple topological descriptors related to nine possible hydroxyl positions on the flavonol/flavone main structure were considered. Multiple linear regression (MLR) modeling technique was employed along with leave-one-out cross-validation (LOO-CV) to ensure the robustness and predictive accuracy of the model.

2.5. Absorption, distribution, metabolism and excretion (ADME)

Finally, the Adsorption, Digestion, Metabolism and Excretion (ADME) study of the shortlisted compounds was predicted using SwissADME server [65].

3. Results and discussion

3.1. Molecular docking of the training set

SARS-CoV-2 Mpro plays a critical role in viral replication, making it a

significant target for inhibiting COVID-19 [66]. Table 1 presents the 50 screened flavone and flavonol compounds, ranked by their AutoDock Vina binding energy scores against SARS-CoV-2 Mpro. The table also provides the ChEMBL IDs for each compound and a binary representation (1 or 0) indicating the presence or absence of hydroxyl groups at nine possible positions (R3, R5, R6, R7, R8, R2', R3', R4', and R5') on the flavone/flavonol scaffold, along with the antiviral drug Nirmatrelvir. Table 2 summarizes the molecular docking results for the top 5 ranked structures, providing insights into their binding affinity and interactions with SARS-CoV-2 Mpro. The table presents each compound's binding score, details of key residue interactions, and visualizations of the hydrogen bond and hydrophobic pockets involved. Notably, the surfaces are color-coded to represent interaction strength: for hydrogen bonding, donor character increases from green to pink, while for hydrophobicity, the character increases from blue to brown. This visual representation allows for a rapid assessment of the key binding features of each compound. As shown in Table 3, the interactions between the protein and ligand in each docking complex are represented by colored dotted lines: light pink for alkyl/pi-alkyl, green for hydrogen bonds, light green for van der Waals and carbon-hydrogen interactions, brick red for pi-carbon or pi-sulfur, parrot green for pi-lon pair, violet for pi-sigma, and dark pink for pi-pi stacked interactions. All docked structures and Nirmatrelvir bound to the active site of SARS-CoV-2 Mpro, engaging residues Thr26, Tyr54, Phe140, Asn142, Gly143, Ser144, Cys145, His163, His164, Glu166, and His172 [58]. Notably, His41 and Cys145, which form the catalytic dyad of Mpro, were the most frequently interacting residues across the top-ranked compounds (Table 3), highlighting their crucial role in ligand binding and, consequently, for Mpro inhibition

Table 2
SARS-CoV-2 Mpro Protein with the top 5 polyphenols interactions: residues, the hydrogen and hydrophobic pockets.

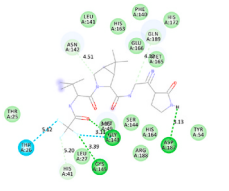
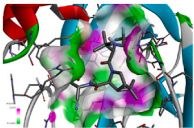
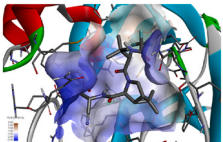
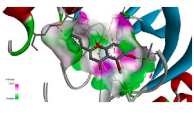
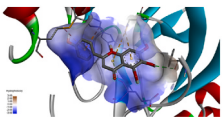
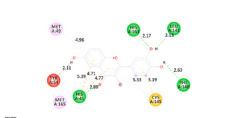
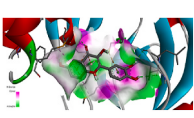
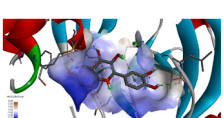
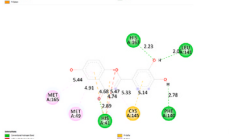
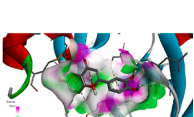
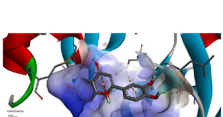
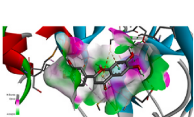
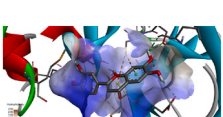
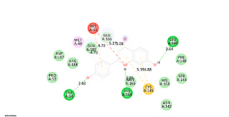
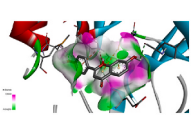
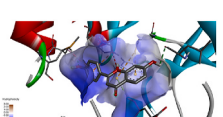
Compound	Score	Rank	Protein Ligand Interaction 2D	Hydrogen Bond Receptor Surface	Hydrophobic Receptor Surface
Nirmatrelvir	−8.7	1			
5,6,7,4'-Tetrahydroxyflavone (6-hydroxyapigenin) F1	−8.3	2			
3,6,3',4'-Tetrahydroxyflavone F2	−8.2	3			
6,3',4'-Trihydroxyflavone F3	−8.2	4			
6,7,3',4'-Tetrahydroxyflavone F4	−8.2	5			
7,4'-Dihydroxyflavone F5	−8.1	6			

Table 3
The top 5 ranked compounds and their type of interactions with SARS-CoV-2 Mpro protein.

Interaction Type by Color	GLU 166	LEU 141	HIS 164	HIS 163	GLN 189	SER 144	PHE 140	HIS 41	ASP 187	CYS 145	MET 165	TYR 54	ARG 188	Met 49
5,6,7,4'-Tetrahydroxyflavone (F1)	+		+				+	+		+				+
3,6,3',4'-Tetrahydroxyflavone (F2)		+		+			+	+		+	+	+		+
6,3',4'-Trihydroxyflavone (F3)		+		+			+	+		+	+			+
6,7,3',4'-Tetrahydroxyflavone (F4)	+	+	+	+		+		+		+	+	+		+
7,4'-Dihydroxyflavone (F5)	+	+	+					+		+		+		+

[67]. This observation is consistent with previous studies emphasizing the importance of these residues for Mpro inhibitor activity [68]. Compound F1 (−8.3 kcal/mol) formed three hydrogen bonds with Mpro through residues Phe140, His164, and Glu166, and exhibited a hydrophobic interaction with Met49, His41 and Cys145. Notably, Glu166 has been implicated as a key residue in Mpro dimerization [37]. Disrupting dimerization could potentially enhance the inhibitory activity of F1 [69]. The F2/Mpro complex (−8.2 kcal/mol) exhibited four hydrogen bonds (His41, Phe140-OH4', Leu141-OH3', and His163-OH3') and the hydrophobic interactions Met49, Met165, His41 and Cys145. However, an unfavorable positive-positive interaction with Tyr54 was also observed, which could potentially reduce binding affinity. Interestingly, the F3/Mpro complex (−8.2 kcal/mol), while sharing the same hydrogen bond and hydrophobic interaction patterns as F2/Mpro, did not exhibit this unfavorable interaction. This suggests that the presence of the OH3 group in F2 significantly influences its interactions with Mpro, potentially leading to the unfavorable positive-positive interaction with Tyr54. The remaining top-ranked compounds, F4 and F5, exhibited distinct binding interactions and affinities. F4, with a binding

energy of −8.2 kcal/mol, formed hydrogen bonds with His164, His163, Ser144, Leu141, and Glu166, and interacted with Tyr54, Met165, Met49, His41, and Cys145. F5, exhibiting a slightly weaker binding energy of −8.1 kcal/mol, formed hydrogen bonds with Leu141, His164, Tyr54, and Glu166, and interacted with Cys145, Met49, and His41. These variations in binding interactions highlight the structural diversity within this set of flavonoid compounds and their potential to engage Mpro in different ways. Fig. 2 illustrates the frequency of hydrogen bonding interactions between specific hydroxyl groups on the fifty flavone/flavonol structures and SARS-CoV-2 Mpro residues. The hydroxyls OH7 and OH3' formed hydrogen bonds 25 and 24 times, respectively, followed by OH4' (17 times), OH6 (11 times), OH8 (8 times), OH2' (7 times), OH3 (6 times), OH5' (4 times), and OH5 (twice). Supplementary Table S1 provides detailed training set molecular docking results, including ligand-protein interactions, with interactive visualizations accessible through the COVID-19 Docking Server.

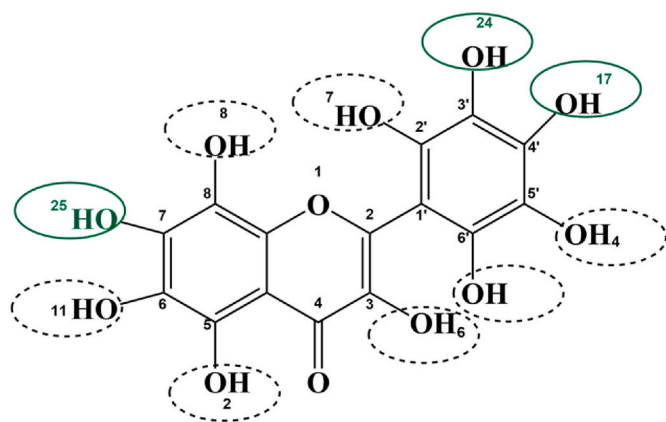


Fig. 2. Number of H-bonds made by the hydroxyls of the selected compound series.

3.2. QSAR modeling

Inverse-QSAR requires a QSAR model built using reversible molecular descriptors. This study employed simple, binary descriptors based on the presence or absence of a hydroxyl group at nine possible positions (R1 to R9) on the flavonol/flavone scaffold. A value of 1 was assigned if a hydroxyl group was present at a given position and 0 if absent.

A QSAR model was developed to predict the AutoDock Vina binding energy score (PDB: 6LU7) as a function of these nine hydroxyl-based descriptors. The model was built using multiple linear regression (MLR) and validated with leave-one-out cross-validation (LOO-CV). The resulting model is presented in Equation (1):

$$\text{Binding Energy Score} = -0.016 \times R3 + 0.0157 \times R5 - 0.417 \times R6 - 0.3375 \times R7 + 0.2282 \times R8 - 0.082 \times R2' - 0.1508 \times R3' - 0.3937 \times R4' + 0.2271 \times R5' - 7.0624 \quad (1)$$

$$(n = 50; R = 0.894; R^2 = 0.7991; s = 0.179)$$

Where n represents the number of compounds, R is the correlation coefficient, R^2 is the coefficient of determination, and s is the standard error of the regression.

The correlation matrix for the nine descriptors (Table 4) indicates low intercorrelation, confirming their linear independence. These descriptors, therefore, can effectively model the AutoDock Vina binding energy score (PDB: 6LU7) for flavonol/flavone compounds. Fig. 3 presents the observed versus calculated binding energy scores, while Fig. 4 shows the observed versus residual scores, visually depicting the model's performance.

3.3. Inverse-QSAR analysis

$2^9 = 512$ hydroxyl combinations were generated and evaluated using the developed inverse-QSAR model. The top 30 predicted compounds with the most favorable binding energy scores are presented in Table 5.

Table 4

Correlation matrix for the nine hydroxyl-based descriptors.

	R3	R5	R6	R7	R8	R2'	R3'	R4'	R5'
R3	1								
R5	0.132	1							
R6	0.105	0.298	1						
R7	0.023	0.274	0.285	1					
R8	0.062	0.062	0.214	0.372	1				
R2'	0.236	0.218	0.179	0.184	0.013	1			
R3'	0.083	0.167	0.048	0.040	0.058	0.393	1		
R4'	0.137	0.053	0.027	0.157	0.074	0.332	0.240	1	
R5'	0.108	0.021	0.048	0.094	0.149	0.107	0.246	0.232	1

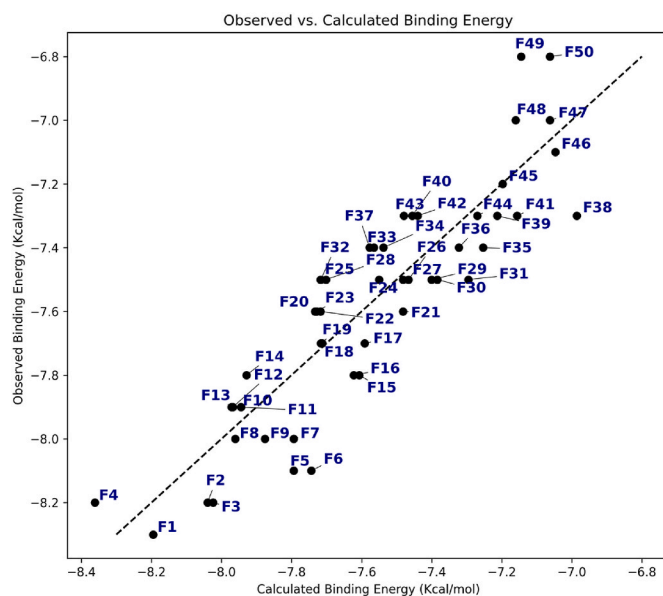


Fig. 3. Observed versus calculated binding energy scores for the QSAR model (Eq. (1)).

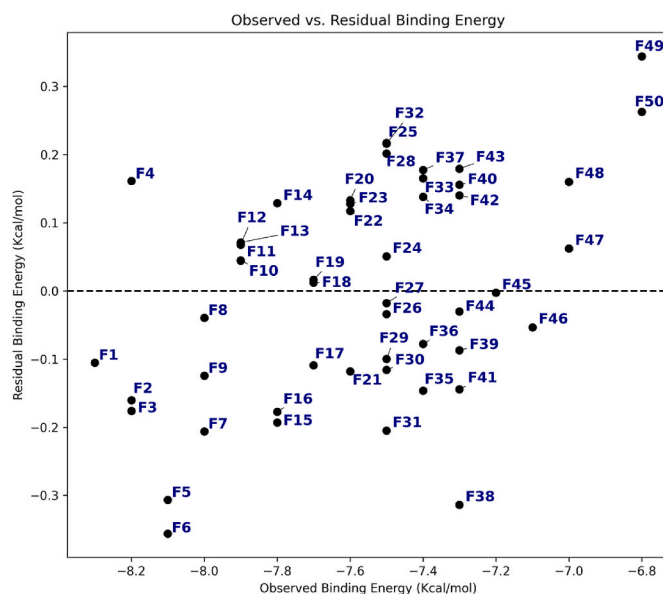


Fig. 4. Observed versus residual binding energy scores for the QSAR model (Eq. (1)).

Detailed molecular docking results for these compounds are provided in Supplementary Table S2, including interactive visualizations from the COVID-19 Docking Server. Notably, three hydroxyl combinations were

Table 5
Predicted binding energy scores in kcal/mol for the top 30 Inverse-QSAR designed flavonol/flavone compounds.

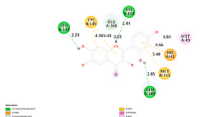
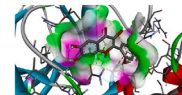
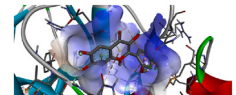
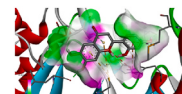
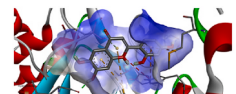
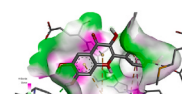
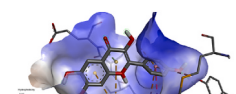
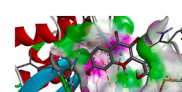
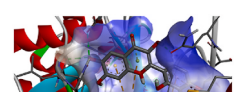
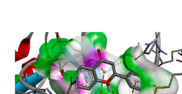
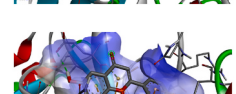
Structure	Nomenclature	R3	R5	R6	R7	R8	R2'	R3'	R4'	R5'	Inverse-QSAR Predicted Score	Binding Energy Score
PF1	3,6,7,2',3',4'-Hexahydroxyflavone	1	0	1	1	0	1	1	1	0	-8,4594	-8.40
PF2		1	1	1	1	0	1	1	1	0	-8,4437	-8.30
PF3		0	0	1	1	0	1	1	1	0	-8,4434	-8.30
PF4		0	1	1	1	0	1	1	1	0	-8,4277	-8.20
PF5		1	0	1	1	0	0	1	1	0	-8,3774	-8.40
PF6		1	1	1	1	0	0	1	1	0	-8,3617	-8.20
PF7		0	0	1	1	0	0	1	1	0	-8,3614	-8.20
PF8	3,6,7,2'4'-Pentahydroxyflavone	0	1	1	1	0	0	1	1	0	-8,3457	-8.20
PF9		1	0	1	1	0	1	0	1	0	-8,3086	-8.50
PF10		1	1	1	1	0	1	0	1	0	-8,2929	-8.40
PF11	6,7,2',4'-Tetrahydroxyflavone	0	0	1	1	0	1	0	1	0	-8,2926	-8.50
PF12		0	1	1	1	0	1	0	1	0	-8,2769	-8.30
PF13		1	0	1	1	0	1	1	1	1	-8,2323	-7.90
PF14	3,6,7,4'-Tetrahydroxyflavone	1	0	1	1	1	1	1	1	0	-8,2312	-7.90
PF15		1	0	1	1	0	0	0	1	0	-8,2266	-8.50
PF16		1	1	1	1	0	1	1	1	1	-8,2166	-7.70
PF17		0	0	1	1	0	1	1	1	1	-8,2163	-8.10
PF18		1	1	1	1	1	1	1	1	0	-8,2155	-8.00
PF19		0	0	1	1	1	1	1	1	0	-8,2152	-7.90
PF20		1	1	1	1	0	0	0	1	0	-8,2109	-8.30
PF21	6,7,4'-Trihydroxyflavone	0	0	1	1	0	0	0	1	0	-8,2106	-8.40
PF22		0	1	1	1	0	1	1	1	1	-8,2006	-7.90
PF23		0	1	1	1	1	1	1	1	0	-8,1995	-8.00
PF24		0	1	1	1	0	0	0	1	0	-8,1949	-8.30
PF25		1	0	1	1	0	0	1	1	1	-8,1503	-7.90
PF26		1	0	1	1	1	0	1	1	0	-8,1492	-7.90
PF27		1	1	1	1	0	0	1	1	1	-8,1346	-7.70
PF28		0	0	1	1	0	0	1	1	1	-8,1343	-7.90
PF29		1	1	1	1	1	0	1	1	0	-8,1335	-8.10
PF30		0	0	1	1	1	0	1	1	0	-8,1332	-8.00

predicted to have more favorable AutoDock Vina binding energy scores (−8.5 kcal/mol) than all compounds in the training set and the reference standard: 3,6,7,2',4'-pentahydroxyflavone, 6,7,2',4'-tetrahydroxyflavone, and 3,6,7,4'-tetrahydroxyflavone.

3.4. Molecular docking analysis of the predicted structures

Molecular docking analysis was performed on the predicted compounds to assess their binding modes and interactions with SARS-CoV-2 Mpro. All predicted compounds docked within the active site of Mpro,

Table 6
SARS-CoV-2 Mpro protein with the top 5 predicted flavonol/flavone interactions: residues, the hydrogen and hydrophobic pockets.

Compound	Score	Rank	Protein Ligand Interaction 2D	Hydrogen Bond Receptor Surface	Hydrophobic Receptor Surface
3,6,7,2',4'-Pentamethoxyflavone PF9	−8.5	1			
6,7,2',4'-Tetramethoxyflavone PF11	−8.5	2			
3,6,7,4'-Tetramethoxyflavone PF15	−8.5	3			
3,6,7,2',3',4'-Hexamethoxyflavone PF1	−8.4	4			
6,7,4'-Trimethoxyflavone PF21	−8.4	5			

engaging similar residues as observed for the training set compounds (Thr26, Tyr54, Phe140, Asn142, Gly143, Ser144, Cys145, His163, His164, Glu166, and His172) [58]. Table 6 provides a detailed overview of the hydrogen bond and hydrophobic pockets involved with the predicted flavonol/flavone. The surfaces are color-coded to indicate interaction strength: green to pink for increasing hydrogen bond donor character and blue to brown for increasing hydrophobicity, enabling rapid visual assessment of key binding features. Table 7 summarizes the types of interactions observed between the predicted compounds (PF1, PF9, PF11, PF15, and PF21) and Mpro. These interactions are visually represented in and color-coded to distinguish the different types: light pink (alkyl/pi-alkyl), green (hydrogen bonds), light green (van der Waals/carbon-hydrogen), brick red (pi-carbon/pi-sulfur), parrot green (pi-lone pair), violet (pi-sigma), and dark pink (pi-pi stacked). Similar to the training set, His41 and Cys145, which constitute the catalytic dyad, emerged as key interacting residues [67], highlighting their importance for ligand binding. Notably, PF9/Mpro complex, with a binding energy of −8.5 kcal/mol, exhibited four hydrogen bonds involving Glu166, Leu141, His164 and Gln 189, along with four hydrophobic interactions with His41, Met49, Cys145 and Met165. The PF11/Mpro complex (−8.5 kcal/mol) showed a similar binding pattern, with four hydrogen bonds (Glu166, His164, Leu141-OH7, and Asp187-OH4') and three hydrophobic interactions with Cys145, Met49 and His41. Likewise, the PF15/Mpro complex (−8.5 kcal/mol) formed four hydrogen bonds Glu166-OH6, Leu141-OH7, His164 and Tyr54-OH4' in addition to four

hydrophobic interactions (His41, Cys145, Met49 and Met165). The PF1/Mpro complex (−8.4 kcal/mol), however, exhibited a more extensive hydrogen bonding network, forming four hydrogen bonds with residues Gln189-OH3, Phe140-OH6, Glu166-OH6, and His164 this complex also showed four hydrophobic interactions with His41, Met49, Met165 and Cys145. Finally, the PF21/Mpro complex, with a binding energy of −8.4 kcal/mol, formed two hydrogen bonds (Glu166-OH6 and His164) and three hydrophobic interactions with His41, Met49 and Cys145. The predicted compounds exhibited diverse hydrogen bonding patterns, suggesting that multiple hydroxyl groups on the flavone/-flavonol scaffold can contribute to Mpro binding. The consistent involvement of Glu166 in hydrogen bonding across multiple compounds suggests this residue is a particularly important interaction point for this class of inhibitors [69]. However, two unfavorable positive-positive interactions with His41 were also observed for PF11 and PF21, which could potentially reduce their binding affinity. The molecular docking analysis supports the inverse-QSAR predictions, showing that the predicted compounds with the most favorable binding energies also exhibit extensive and favorable interactions with key Mpro residues. Supplementary Table S2 provides detailed predicted molecular docking results, including ligand-protein interactions, with interactive visualizations accessible through the COVID-19 Docking Server links.

Table 7
The top 5 predicted compounds and their type of interactions with SARS-CoV-2 Mpro protein.

Interaction Type by Color		Van der Waals		Conventional Hydrogen Bond				Pi-Alkyl		Carbon Hydrogen Bond		Unfavorable Positive-Positive		Pi-Sulfur
Residue	GLU 166	LEU 141	HIS 164	HIS 163	GLN 189	SER 144	PHE 140	HIS 41	ASP 187	CYS 145	MET 165	TYR 54	ARG 188	Met 49
3,6,7,2'4'-Pentahydroxyflavone (PF9)	+	+	+		+			+		+	+			+
6,7,2',4'-Tetrahydroxyflavone (PF11)	+	+	+					+	+	+				+
3,6,7,4'-Tetrahydroxyflavone (PF15)	+	+	+					+		+	+	+		+
3,6,7,2',3',4'-Hexahydroxyflavone (PF1)	+		+		+		+	+		+	+			+
6,7,4'-Trihydroxyflavone (PF21)	+		+					+		+				+

3.5. MD simulation analysis

To further evaluate the stability of the predicted PF9/Mpro complex, a 300 ns molecular dynamics (MD) simulation was performed. This simulation aimed to analyze conformational fluctuations of the complex over time. The root mean square deviation (RMSD) is a measure of structural stability, quantifying the average deviation of atom positions from a reference structure. Fig. 5a presents the RMSD of the PF9/Mpro complex over the simulation time. The RMSD values fluctuated but remained ≤ 3.6 Å, consistent with the range observed for other phytochemical complexes with 6LU7 [68], indicating that the PF9/Mpro complex remained stable throughout the 300 ns simulation.

The root mean square fluctuation (RMSF) measures the flexibility of individual residues, quantifying their average positional fluctuations over time. The RMSF plot (Fig. 5b) reveals that residues Leu141 (3.6 Å), Asn142 (3.2 Å), Tyr154 (3.7 Å), Arg 222 (3.7 Å), Asn 277 (3.4 Å), and Gly 278 (3.1 Å) exhibited the highest fluctuations. Importantly, except for Leu141 and Asn142, these residues are located outside the binding pocket, and the catalytic dyad (His41 and Cys145) remained stable [67]. This suggests that these fluctuations did not significantly impact the binding interactions between PF9 and Mpro. This observation is supported by the average of two hydrogen bonds maintained between PF9 and Mpro throughout the simulation (Fig. 5c).

The radius of gyration (Rg) assesses the compactness of the protein structure during the simulation. Fig. 5d shows the Rg of the PF9/Mpro complex, which remained steady throughout the simulation. This consistent Rg value suggests that the complex maintained its compactness and stability [70].

The solvent-accessible surface area (SASA) quantifies the surface area of a biomolecule accessible to a solvent. Changes in SASA can provide insights into a complex's interaction with its environment, stability, folding, and binding properties. Fig. 5e presents the SASA of

the PF9/Mpro complex. The observed decrease in SASA after 170 ns suggests that the complex became more compact, potentially due to conformational changes that buried more of its surface area. This further supports the conclusion that the PF9/Mpro complex is stable.

Taken together, the RMSD, RMSF, Rg, and SASA analyses provide strong evidence for the stability of the PF9/Mpro complex over the 300 ns MD simulation.

3.6. ADME

The ADME properties of the compounds from the training dataset (denoted as F) and the inverse-QSAR predicted compounds (denoted as PF) offer valuable insights into their potential as drug candidates (Table 8). The molecular weights (MW) of these compounds range from 254.24 to 318.24, which is favorable for oral bioavailability. Their cLogP values around 0.55 indicate balanced lipophilicity, essential for membrane permeability and absorption. The hydrogen bond acceptors (HBA) and donors (HBD) numbers generally align with Lipinski's rule of five, suggesting good oral bioavailability.

Topological polar surface area (TPSA) values range from 70.67 Å² to 151.59 Å², with lower TPSA typically favoring better cell membrane permeability. Skin permeation values (LogKp) from -7.44 to -5.54 cm/s indicate limited skin permeability. Solubility varies from soluble to moderately soluble, with soluble compounds being more bioavailable orally. All compounds exhibit zero or one Lipinski violation, indicating potential oral activity.

The predicted compounds generally show similar or slightly better ADME properties compared to the training set compounds, with high solubility and minimal Lipinski violations, making them promising candidates for further development. The overall balanced lipophilicity, favorable solubility, adherence to Lipinski's rules, and appropriate TPSA values suggest good oral bioavailability and drug-likeness, essential for

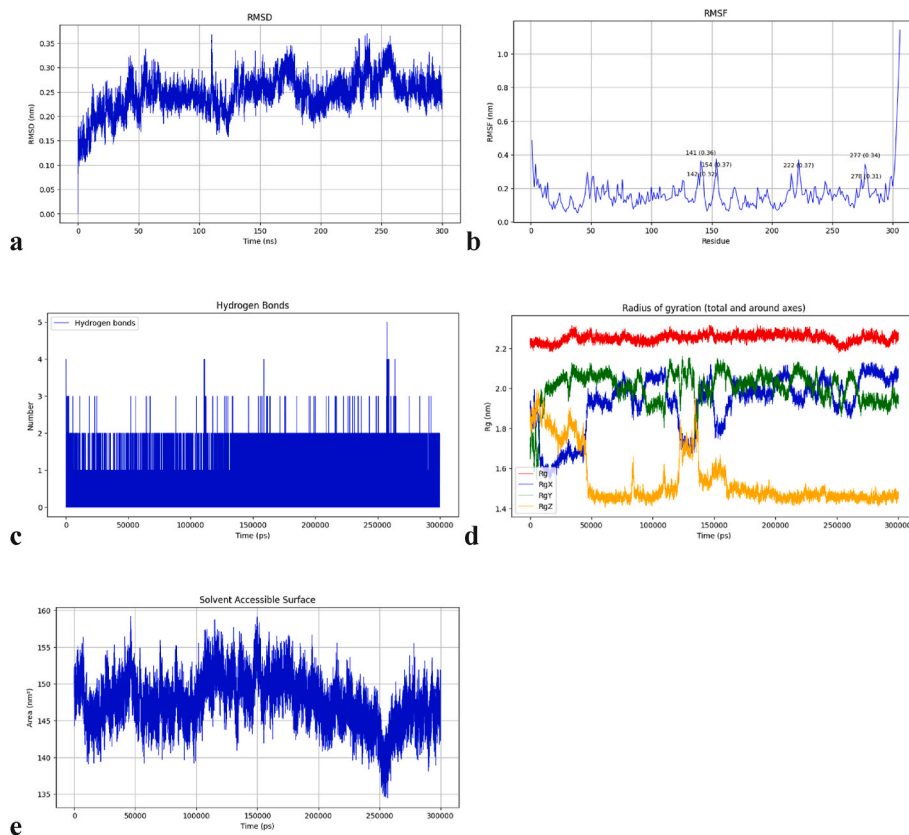


Fig. 5. Molecular Dynamics Simulation Analysis of the PF9/Mpro Complex (300 ns). (a) RMSD vs. time. (b) RMSF vs. time. (c) Number of hydrogen bonds between PF9 and Mpro vs. time. (d) Radius of gyration vs. time. (e) Solvent-accessible surface area (SASA) vs. time.

Table 8

ADME properties of the top ranked compounds from the training data set and the inverse-QSAR predicted compounds.

Compound	Oral bioavailability							Pharmacokinetic properties	LogKp (Skin permeation) (cm/s)	Ali solubility Class [72]	Lipinski Violation)
	MW	cLogP	HBA	HBD	RB	TPSA (Å ²)	B-Score				
PF9	302.24	1.06	7	5	1	131.36	0.55	High	−7.44	Soluble	0
PF11	286.24	1.48	6	4	1	111.13	0.55	High	−6.8	Soluble	0
PF15	286.24	1.57	6	4	1	111.13	0.55	High	−6.65	Soluble	0
PF1	318.24	0.85	8	6	1	151.59	0.55	Low	−7.35	Moderately soluble	1
PF21	270.24	2.02	5	3	1	90.9	0.55	High	−5.88	Moderately soluble	0
F1	286.24	1.81	6	4	1	111.13	0.55	High	−6.16	Moderately soluble	0
F2	288.25	1.53	6	5	3	118.22	0.55	High	−6.16	Moderately soluble	0
F3	270.24	2.05	5	3	1	90.9	0.55	High	−5.88	Moderately soluble	0
F4	286.24	1.55	6	4	1	111.13	0.55	High	−6.64	Soluble	0
F5	254.24	2.4	4	2	1	70.67	0.55	High	−5.54	Moderately soluble	0

effective therapeutic agents.

While these predictions provide valuable information, it's important to acknowledge the limitations of *in silico* ADME predictions [71]. Experimental validation is crucial to confirm these properties and assess their overall suitability for drug development.

4. Conclusion

This study introduces a novel methodology for *de novo* drug design by integrating inverse-QSAR and molecular docking. Utilizing simple, reversible descriptors, this approach overcomes limitations of traditional inverse-QSAR techniques, enabling efficient generation of novel scaffold-based structures with predicted activity. This method's versatility, applicable to various protein targets and chemical scaffolds, offers a powerful platform for advancing drug discovery efforts.

To demonstrate the method's potential, we conducted a case study targeting the SARS-CoV-2 main protease (Mpro), a key enzyme for viral replication. A QSAR model was developed using a dataset of 50 flavone and flavonol compounds and their calculated binding affinities to Mpro. Through inverse-QSAR analysis of 2⁹ = 512 possible hydroxyl combinations, we identified three novel compounds predicted to exhibit high binding affinities to Mpro: 3,6,7,2',4'-pentahydroxyflavone, 6,7,2',4'-tetrahydroxyflavone, and 3,6,7,4'-tetrahydroxyflavone. Molecular dynamics simulations further demonstrated the stability of the PF9/Mpro complex, supporting its potential as a lead compound. These findings highlight the ability of this combined computational approach to identify novel and promising inhibitors, paving the way for the development of new therapeutics against SARS-CoV-2 and potentially other diseases.

Further applications of this methodology to other targets and scaffolds are currently under investigation.

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Conflicts of interest/Competing interests

The authors have no relevant financial or non-financial interests to disclose.

Availability of data and material

All data generated or analyzed during this study are included in this published article.

Code availability

N.A.

CRediT authorship contribution statement

Rafik Menacer: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Saad Boucheikioua:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Investigation. **Saida Meliani:** Writing – review & editing, Formal analysis, Data curation. **Nadjah Belattar:** Data curation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used CHATGPT and Google AI Studio in order to improve readability and language of some parts of the manuscript. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The authors dedicate this paper to the memory of the brilliant Professor Mebarki Nouredine who sadly passed away on Monday 20th September 2021.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compbimed.2024.108992>.

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